

Ashwin Satish

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EDUCATION

Northeastern University

Master's of Science, Data Science (GPA: 3.83)

Machine Learning, Algorithms, Data Mining, NLP, MLOps, Capstone

Boston, MA

Sept 2024 - Dec 2026

Christ University

Bachelor's of Science, Computer Science (GPA: 3.53)

Bangalore, India

June 2020 - May 2023

EXPERIENCE

Gyori Lab for Computational Biomedicine - Machine Learning Intern

Jan 2026 - Present | Boston, MA

- Parsed **35M** PubMed records, identifying **278K+ publications** linked to registered clinical trials & built [TrialSynth](#)'s end-to-end clinical trial LLM extraction pipeline, turning **13K+ publications** into structured trial datasets for knowledge graph integration.
- Designed & implemented [CoGEx](#)'s **Trial Result** schema, converting trial datasets to **7 Neo4j node** types & **9 relationship** types, generating **115K+ nodes** & **116K+ relationships**, enabling queryable clinical trials within the knowledge graph.
- Boosted adverse event entity coverage from **32.5% to 66.8%** across **13k+** clinical trial papers by building a multi-ontology extraction pipeline with systematic noise filtering.
- Delivered a **production Clinical Trial Browser** ([discovery.indra.bio](#)) with 13 REST APIs & a Flask search interface, enabling real-time lookup of trial outcomes across the CoGEx biomedical knowledge graph.

Northeastern University - Machine Learning Researcher

Jan 2025 - Present | Boston, MA

- Boosted true positives from **2% to 77%** while **maintaining zero false positives** for early-stage Lyme cell detection via a 7-model CNN ensemble (TensorFlow).
- Built a pipeline for reconstruction & masking of **10,000+ cell images**, saving the biology team weeks of manual work.
- Improved accuracy by **13%** via augmentation & L2/dropout regularization, achieving production-ready performance.

Northeastern University - Teaching Assistant (Foundations of Data Science)

Aug 2025 - Dec 2025 | Boston, MA

- Mentored **300+ students** in linear algebra & statistics for ML & Data Science via problem-solving sessions & **NumPy** demos.
- Conducted **4+ weekly** office hours clarifying topics like constraint optimization, low-rank approximation, & regression.

CarerAI - Data Science Intern

Jul 2023 - May 2024 | Gurgaon, India

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through KNN analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's accuracy by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

SKILLS

Programming | Data: Python, SQL, R, Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV

Visualization | Cloud: Git, Tableau, Matplotlib, Seaborn, AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j

AI | Modeling: Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Kubeflow, Ollama, PydanticAI, MLFlow

PROJECTS

[AI-Powered Pokémon Battle Strategy Engine](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating **Llama 3.2** with **deterministic backend** for strategy analysis with a recommendation system for optimal team synergies by ingesting **1M+ aggregated usage stats** via an ETL pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, **indexing 40,000+ strategy docs** to ground LLM inference in semantic context.

[PantryPilot \(2nd Place | MLOps Innovation Expo @ Google\)](#)

- Contributed to the fine tuning, evaluation & automated deployment (**GCP**) of **Llama 3.2 3B** for personalized inventory recipes, resulting in **50% reduction** in **dietary violations** via LoRA adaptation.
- Collaborated on DPO feedback with Teacher model (**Llama 3 70B**) for preference labelling, VLM for OCR & Veo 3.1 for video.